Business Intelligence – Exercise Sheet: Perceptron Calculations by Hand

Bachelor in Wirtschaftsinformatik, 6th Semester

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Exercises on Feed-Forward and Backpropagation in Perceptrons

Use column vectors throughout. For every forward pass, compute all intermediate net inputs $z^{(\ell)}$ and activations $a^{(\ell)}$. Unless stated otherwise, use

$$a^{(0)} = x, \qquad z^{(\ell)} = W_\ell a^{(\ell-1)} + b_\ell.$$

Use

$$\mathrm{ReLU}(z) = \max(0, z), \qquad \sigma(z) = \frac{1}{1+e^{-z}}, \qquad \mathrm{softmax}(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}.$$

For backpropagation tasks, use gradient descent

$$\theta_{\text{new}} = \theta_{\text{old}} - \eta \nabla_\theta L.$$

When numerical approximations are needed, round to three decimals. For $z = 0$, use $\mathrm{ReLU}'(0) = 0$. Leave softmax results in exact exponential form whenever convenient.

1. **Ex 1 (20 Points):** A simple invoice screening perceptron uses one neuron.

   (a) Given

   $$x = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}, \qquad w = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}, \qquad b = 1,$$

   compute the net input

   $$z = w^\top x + b.$$

   (8 Points)

(b) Use the decision rule

$$\hat{y} = \begin{cases} 1, & z \geq 0, \\ 0, & z < 0. \end{cases}$$

Determine the predicted class. (4 Points)

(c) Repeat the calculation for

$$x' = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

using the same w and b. (8 Points)

2. **Ex 2 (20 Points):** A lead scoring model uses one layer with two output neurons.

(a) Given

$$x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \qquad W_1 = \begin{bmatrix} 1 & 0 & -1 \\ 2 & -1 & 1 \end{bmatrix}, \qquad b_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

compute

$$z^{(1)} = W_1 x + b_1.$$

(8 Points)

(b) Apply ReLU to obtain

$$a^{(1)} = \mathrm{ReLU}(z^{(1)}).$$

(6 Points)

(c) Which output neuron is selected if the decision rule is "choose the neuron with the largest activation"? (6 Points)

3. **Ex 3 (20 Points):** A credit pre-check model uses one hidden layer and one output neuron.

(a) Given

$$x = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \qquad W_1 = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 0 & 2 \end{bmatrix}, \qquad b_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

compute

$$z^{(1)} = W_1 x + b_1, \qquad a^{(1)} = \mathrm{ReLU}(z^{(1)}).$$

(6 Points)

(b) Given

$$W_2 = \begin{bmatrix} 2 & -1 \end{bmatrix}, \qquad b_2 = -1,$$

compute the output score

$$z^{(2)} = W_2 a^{(1)} + b_2.$$

(8 Points)

(c) Use the decision rule

$$\hat{y} = \begin{cases} 1, & z^{(2)} \geq 0, \\ 0, & z^{(2)} < 0. \end{cases}$$

Determine the predicted class. (6 Points)

4. **Ex 4 (20 Points):** A recommender system assigns a customer to one of two product categories.

(a) Given

$$x = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \qquad W_1 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 0 \end{bmatrix}, \qquad b_1 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix},$$

compute the hidden-layer values

$$z^{(1)} = W_1 x + b_1, \qquad a^{(1)} = \mathrm{ReLU}(z^{(1)}),$$

and identify which hidden neurons are active. (6 Points)

(b) Given

$$W_2 = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 2 & 0 \end{bmatrix}, \qquad b_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

compute the output scores

$$z^{(2)} = W_2 a^{(1)} + b_2.$$

(8 Points)

(c) Determine the predicted class using the rule

$$\hat{y} = \arg\max_i z_i^{(2)}.$$

(6 Points)

5. **Ex 5 (20 Points):** A churn model uses one sigmoid output neuron.

   (a) Given

   $$x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad w = \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \qquad b = -1,$$

   compute

   $$z = w^\top x + b.$$

   (6 Points)

   (b) Compute the predicted probability

   $$\hat{y} = \sigma(z).$$

   (8 Points)

   (c) Determine the predicted class once with threshold 0.5 and once with threshold 0.8. (6 Points)

6. **Ex 6 (20 Points):** A marketing model classifies a customer into one of three segments with SoftMax.

   (a) Given

   $$x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad W_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \end{bmatrix}, \qquad b_1 = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix},$$

   compute the logits

   $$z^{(1)} = W_1 x + b_1.$$

   (6 Points)

   (b) Compute the SoftMax output

   $$\hat{y} = \mathrm{softmax}(z^{(1)}).$$

   Give the result in exact form and, if you wish, also as decimals. (10 Points)

   (c) Determine the predicted class and the corresponding confidence. (4 Points)

7. **Ex 7 (20 Points):** A BI pipeline uses two hidden layers before the final decision.

(a) Given

$$x = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \qquad W_1 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \qquad b_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

compute

$$z^{(1)}, \qquad a^{(1)} = \mathrm{ReLU}(z^{(1)}).$$

(5 Points)

(b) Given

$$W_2 = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}, \qquad b_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

compute

$$z^{(2)}, \qquad a^{(2)} = \mathrm{ReLU}(z^{(2)}).$$

(5 Points)

(c) Given

$$W_3 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad b_3 = \begin{bmatrix} 0 \\ -1 \end{bmatrix},$$

compute the output logits

$$z^{(3)} = W_3 a^{(2)} + b_3$$

and the SoftMax prediction

$$\hat{y} = \mathrm{softmax}(z^{(3)}).$$

(6 Points)

(d) Determine the predicted class. (4 Points)

8. **Ex 8 (20 Points):** A single linear neuron is used for a simplified sales estimate.

(a) Given

$$x = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \qquad w = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad b = 0, \qquad y = 3,$$

compute the forward pass

$$\hat{y} = w^\top x + b$$

and the loss

$$L = \frac{1}{2}(\hat{y} - y)^2.$$

(6 Points)

(b) Compute the gradients

$$\frac{\partial L}{\partial w}, \qquad \frac{\partial L}{\partial b}.$$

(8 Points)

(c) Use learning rate

$$\eta = 0.1$$

and compute the updated parameters w_{new} and b_{new}. (6 Points)

9. **Ex 9 (20 Points):** A binary purchase model uses one sigmoid neuron and binary cross-entropy.

(a) Given

$$x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad w = \begin{bmatrix} 0 \\ -1 \end{bmatrix}, \qquad b = 1, \qquad y = 1,$$

compute

$$z = w^\top x + b, \qquad \hat{y} = \sigma(z).$$

(6 Points)

(b) Use binary cross-entropy and the fact

$$\frac{\partial L}{\partial z} = \hat{y} - y.$$

Compute

$$\frac{\partial L}{\partial w}, \qquad \frac{\partial L}{\partial b}.$$

(8 Points)

(c) Use learning rate

$$\eta = 0.1$$

and compute the updated values of w and b. (6 Points)

10. **Ex 10 (20 Points):** A fraud screening network has one hidden layer, but only the output layer is updated.

(a) Given

$$x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad W_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad b_1 = \begin{bmatrix} 0 \\ -1 \end{bmatrix},$$

compute

$$a^{(1)} = \text{ReLU}(W_1 x + b_1).$$

(6 Points)

(b) Given

$$W_2 = \begin{bmatrix} 1 & -1 \end{bmatrix}, \qquad b_2 = 0, \qquad y = 1,$$

compute

$$\hat{y} = W_2 a^{(1)} + b_2$$

and

$$L = \frac{1}{2}(\hat{y} - y)^2.$$

(4 Points)

(c) Keeping W_1 and b_1 fixed, compute the gradients for W_2 and b_2, and update them with

$$\eta = 0.2.$$

(10 Points)

11. **Ex 11 (20 Points):** A demand scoring network uses one hidden ReLU layer and a linear output.

(a) Given

$$x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad W_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad b_1 = \begin{bmatrix} 0 \\ -1 \end{bmatrix},$$

$$W_2 = \begin{bmatrix} 1 & -2 \end{bmatrix}, \qquad b_2 = 1, \qquad y = 1,$$

compute the full forward pass and the loss

$$L = \frac{1}{2}(\hat{y} - y)^2.$$

(5 Points)

(b) Compute the error signals

$$\delta^{(2)} = \frac{\partial L}{\partial z^{(2)}}, \qquad \delta^{(1)} = \frac{\partial L}{\partial z^{(1)}}.$$

(5 Points)

(c) Compute all gradients and update

$$W_2,\ b_2,\ W_1,\ b_1$$

with

$$\eta = 0.1.$$

(10 Points)

12. **Ex 12 (20 Points):** An upsell classifier uses a hidden ReLU layer and a sigmoid output.

    (a) Given

$$x = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \qquad W_1 = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \qquad b_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

$$W_2 = \begin{bmatrix} 1 & -1 \end{bmatrix}, \qquad b_2 = 0, \qquad y = 1,$$

    compute the full forward pass

$$a^{(1)} = \text{ReLU}(z^{(1)}), \qquad \hat{y} = \sigma(z^{(2)}).$$

    (5 Points)

    (b) Using binary cross-entropy, compute the error signals

$$\delta^{(2)} = \hat{y} - y, \qquad \delta^{(1)}.$$

    (7 Points)

    (c) Compute all gradients and update

$$W_2,\ b_2,\ W_1,\ b_1$$

    with

$$\eta = 0.2.$$

    (8 Points)

13. **Ex 13 (20 Points):** A document classifier uses one hidden layer and a SoftMax output layer. Only the output layer is trained in this step.

    (a) Given

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad W_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad b_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

$$W_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad b_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \qquad y = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

    compute

$$a^{(1)} = \text{ReLU}(W_1 x + b_1), \qquad z^{(2)} = W_2 a^{(1)} + b_2, \qquad \hat{y} = \text{softmax}(z^{(2)}).$$

    (6 Points)

(b) Using cross-entropy, compute the output-layer gradient via

$$\delta^{(2)} = \hat{y} - y.$$

Then determine

$$\frac{\partial L}{\partial W_2}, \qquad \frac{\partial L}{\partial b_2}.$$

(8 Points)

(c) Update W_2 and b_2 with

$$\eta = 0.2.$$

(6 Points)

14. **Ex 14 (20 Points):** A supplier classification model uses one hidden ReLU layer and a SoftMax output layer. This time all layers are updated.

(a) Given

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad W_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad b_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

$$W_2 = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \qquad b_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \qquad y = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

compute the full forward pass. (4 Points)

(b) Compute the error signals

$$\delta^{(2)} = \hat{y} - y, \qquad \delta^{(1)}.$$

(8 Points)

(c) Compute all gradients and update

$$W_2,\ b_2,\ W_1,\ b_1$$

with

$$\eta = 0.1.$$

(8 Points)

15. **Ex 15 (20 Points):** A model is trained online with two consecutive stochastic gradient descent steps.

(a) Start with

$$W = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \qquad b = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \qquad \eta = 1.$$

For the first training example

$$x^{(A)} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad y^{(A)} = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

compute the SoftMax output, the gradient, and the updated W and b. (8 Points)

(b) Using the updated parameters from part (a), perform the forward pass for the second training example

$$x^{(B)} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad y^{(B)} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

(4 Points)

(c) Perform the second SGD update and give the final values of W and b. (8 Points)

16. **Ex 16 (20 Points):** A model is trained with a mini-batch of two examples instead of online updates.

(a) Start with

$$W = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \qquad b = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \qquad \eta = 1.$$

For the two training examples

$$x^{(1)} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad y^{(1)} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad x^{(2)} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad y^{(2)} = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

compute the individual output-layer gradients for sample 1 and sample 2. (8 Points)

(b) Compute the average mini-batch gradient for W and b. (6 Points)

(c) Update W and b once using the average gradient. (6 Points)

17. **Ex 17 (20 Points):** A credit scoring network contains a dead ReLU neuron.

(a) Given

$$x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad W_1 = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad b_1 = \begin{bmatrix} -1 \\ -1 \end{bmatrix},$$

compute

$$z^{(1)}, \qquad a^{(1)} = \mathrm{ReLU}(z^{(1)}),$$

and identify which hidden neuron is inactive. (6 Points)

(b) Given

$$W_2 = \begin{bmatrix} 2 & -1 \end{bmatrix}, \qquad b_2 = 0, \qquad y = 1,$$

compute the output

$$\hat{y} = W_2 a^{(1)} + b_2$$

and the output-layer error signal for the loss

$$L = \frac{1}{2}(\hat{y} - y)^2.$$

(4 Points)

(c) Compute the hidden-layer error signal and show explicitly which entries of

$$\frac{\partial L}{\partial W_1} \quad \text{and} \quad \frac{\partial L}{\partial b_1}$$

are zero because of ReLU. (4 Points)

(d) Update all non-zero parameters with

$$\eta = 0.1.$$

(6 Points)

18. **Ex 18 (20 Points):** A churn model uses L_2-regularization to reduce overfitting.

(a) Given

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad w = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \qquad b = 0, \qquad y = 1, \qquad \lambda = 0.2,$$

compute

$$z = w^\top x + b, \qquad \hat{y} = \sigma(z),$$

and the total loss

$$L = -\big(y \ln(\hat{y}) + (1 - y) \ln(1 - \hat{y})\big) + \frac{\lambda}{2} \|w\|_2^2.$$

Do not regularize the bias. (6 Points)

(b) Using

$$\frac{\partial L}{\partial z} = \hat{y} - y,$$

compute the gradient

$$\frac{\partial L}{\partial w}$$

including the L_2-term, and also compute

$$\frac{\partial L}{\partial b}.$$

(8 Points)

(c) Update w and b with

$$\eta = 0.1.$$

(6 Points)

19. **Ex 19 (20 Points):** A deep classifier uses two hidden ReLU layers and a SoftMax output layer.

(a) Given

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad W_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad b_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

$$W_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad b_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

$$W_3 = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \qquad b_3 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \qquad y = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

compute the complete forward pass up to

$$\hat{y} = \text{softmax}(z^{(3)}).$$

(5 Points)

(b) Compute the error signals

$$\delta^{(3)}, \qquad \delta^{(2)}, \qquad \delta^{(1)}.$$

(7 Points)

(c) Compute all gradients and update

$$W_3,\ b_3,\ W_2,\ b_2,\ W_1,\ b_1$$

with

$$\eta = 0.1.$$

(8 Points)

20. **Ex 20 (20 Points):** Capstone exercise: a campaign response model uses four inputs, two hidden ReLU layers, and a three-class SoftMax output.

(a) Given

$$x = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 2 \end{bmatrix}, \qquad W_1 = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & -1 & 0 \end{bmatrix}, \qquad b_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

$$W_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad b_2 = \begin{bmatrix} -2 \\ 0 \end{bmatrix},$$

$$W_3 = \begin{bmatrix} 1 & -1 \\ -1 & 1 \\ 0 & 0 \end{bmatrix}, \qquad b_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \qquad y = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix},$$

compute the complete forward pass

$$z^{(1)},\ a^{(1)},\ z^{(2)},\ a^{(2)},\ z^{(3)},\ \hat{y}.$$

(6 Points)

(b) Using cross-entropy with SoftMax, compute

$$\delta^{(3)} = \hat{y} - y, \qquad \delta^{(2)}, \qquad \delta^{(1)}.$$

(6 Points)

(c) Compute all gradients and update

$$W_3,\ b_3,\ W_2,\ b_2,\ W_1,\ b_1$$

with

$$\eta = 0.1.$$

(6 Points)

(d) Perform one new forward pass with the updated parameters and decide whether the probability of the correct class has increased. (2 Points)